

# MIDAS GUI — Development History

---

A commit-by-commit record of what changed and when, so you can (a) find **when** a particular behaviour/feature entered the code and (b) know **what to roll back** to undo a specific effect.

Keep this file updated when you add commits: append a new entry at the bottom of the chronological log and add a row to the quick-reference index. Regenerate the PDF the same way as the other docs (pandoc + headless Chrome) if you keep one.

---

## How to use this document

---

- **Find when a change landed:** search the change → commit index below, or `git log --oneline --<path>` for a specific file, or `git log -S"<code string>"` to find the commit that introduced/removed a string.
- **See a commit's exact diff:** `git show <hash>` (whole commit) or `git show <hash> -- <path>` (one file).
- **Undo one commit cleanly (keeps later history):** `git revert <hash>`. For a range: `git revert <oldest>^..<newest>`.
- **Undo just one file back to a commit:** `git checkout <hash> -- <path>` then commit. To see the file *as of* a commit without changing anything: `git show <hash>:<path>`.
- **Hard reset to a point (destructive — local only, never on pushed history):** `git reset --hard <hash>`. Prefer `git revert` on a shared branch.
- **Dependencies matter.** Many later commits build on earlier ones (e.g. the unified data-loader, the config overlay). The "Roll back" note on each entry flags when reverting will disturb dependents.

Branch: `main`. All commits are authored by Dishant Beniwal (AI pair-authored). Dates are commit dates (YYYY-MM-DD).

---

## Change → commit quick-reference index

---

### Packaging / build / launch

| Change                                                              | Commit  |
|---------------------------------------------------------------------|---------|
| Initial v1.0.0 release (9-tab app, all backends)                    | f19ee8c |
| setuptools.build_meta backend (broader pip compat)                  | 3cd1f36 |
| <code>launch.py</code> standalone launcher                          | 2e9f419 |
| Startup crash diagnostics + robust launcher (Windows)               | 916ece2 |
| MIDAS-style packaging (LICENSE, pyproject, release.sh, smoke tests) | e862135 |

## Data Viewer (Tab 0)

| Change                                                                          | Commit  |
|---------------------------------------------------------------------------------|---------|
| Radial-integration plot, beam-centre ring picking, zoom persistence             | acf0866 |
| Zoom/pan preserved across frames (autoRange off)                                | d9e527e |
| Intensity-range mask, calibration-file loading, click-to-draw ring              | d50b588 |
| Compact 3-per-row lattice + cubic quick-set; mask default max(99.99pct,1e5)     | 41f28f5 |
| Projection Skip-frames field; compact geometry/intensity fields; Lsd separators | 108ea7d |
| Intensity-statistics panel + histogram (bottom of loader)                       | 2b7a695 |

## Mask Builder (Tab 1)

| Change                                                                                                | Commit  |
|-------------------------------------------------------------------------------------------------------|---------|
| Unbounded $\sigma$ fields; independent spatial/temporal auto-mask; Viewer↔Calibrate geometry hand-off | 10df828 |
| Temporal-constancy accepts a 3-D HDF5 (time,y,x) stack                                                | 2385f75 |

## Calibrate (Tab 2) / Refinement (Tab 3)

| Change                                                                      | Commit  |
|-----------------------------------------------------------------------------|---------|
| Abort buttons; dark/bright/background field correction; auto-load on browse | 7d010d7 |
| Calibration input accepts paramstest/poni/json                              | 41f28f5 |
| Load-calibration-file (geometry + $\lambda$ ) helpers                       | b381d8d |

## Batch Integrate (Tab 4)

| Change                                                                       | Commit  |
|------------------------------------------------------------------------------|---------|
| Live folder MONITOR + per-file legend                                        | 1149afc |
| Clear results + inline per-file labels & plot controls                       | 524f5f1 |
| Stacked profiles: publication themes, point+line, symbol size, x-units, grid | d135c80 |
| Waterfall R/2 $\theta$ /Q x-axis selector                                    | 2385f75 |

## PDF Analysis (Tab 6)

| Change                                                                             | Commit  |
|------------------------------------------------------------------------------------|---------|
| Polyatomic midas_pdf reduction pipeline + vendored midas_pdf + calibration loading | b381d8d |
| Defaults to real I(Q) test data                                                    | 1149afc |

## Pump Probe (TR-XRD) tab

| Change                                                                                                                                                    | Commit  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| New tab: TRR pooling + MIDAS-engine integration + $\Delta I(q, \text{delay})$ + 4 pyqtgraph views + publication-quality plots + default MIDAS calibration | 590b410 |

## Cross-cutting UI / infrastructure

| Change                                                                        | Commit  |
|-------------------------------------------------------------------------------|---------|
| Linux visible files/folders style fix                                         | fc754d0 |
| Pixel-weighted azimuthal mean; readable QMenu; no-scroll combos               | 41f28f5 |
| Unified Data-Loader panel + draggable 3-panel splitter (tabs 0/2/3/4)         | aa9395e |
| Clickable $\lambda$ label with K-edge foil menu                               | b628446 |
| Clickable px label with detector pixel-size menu                              | 3248638 |
| Per-user JSON configuration system + Preferences dialog                       | d30be2c |
| Modular tab visibility (show/hide optional tabs)                              | 590b410 |
| User-adjustable interface scaling (HiDPI / 4K, QT_SCALE_FACTOR)               | d5fafd8 |
| Lsd displayed in mm (calculations & calibration files stay in $\mu\text{m}$ ) | c4e1c12 |

## Stability / performance / consistency (review-driven, phases 1–3)

| Change                                                                                   | Commit  |
|------------------------------------------------------------------------------------------|---------|
| Clean worker shutdown, drop QThread.terminate(), guard stdout, offload projection        | 399f3d7 |
| Waterfall $O(N^2) \rightarrow O(N)$ , frame-slider debounce, radial-grid + stats caching | 4462ee1 |
| Dedupe distortion map + paramstest writer, align calibrant lattices, texture colormap    | 8846619 |

## Documentation

| Change                                                              | Commit  |
|---------------------------------------------------------------------|---------|
| README not-on-PyPI clone/conda instructions                         | cd05ced |
| gui_documentation.md added                                          | 2e5f7c9 |
| gui_documentation.pdf added                                         | 75c135c |
| README/environment install via <code>pip install midas_suite</code> | aceb40f |
| implementation_details.md/.pdf added                                | 524f5f1 |
| config_gui.md + config.example.json added                           | d30be2c |
| development_history.md/.pdf added                                   | 6d6f3fb |

---

## Chronological commit log (oldest → newest)

---

### **f19ee8c** — Initial release: midas\_gui v1.0.0 (2026-06-30)

**Effect:** First version. Nine-tab PyQt5 GUI over `midas_calibrate_v2` / `midas_integrate_v2`: mask building, auto-calibration, refinement, batch integration + waterfall, corrections, PDF, texture, export. Entry points `midas-gui` / `python -m midas_gui`. **Files:** entire `midas_gui/` package + `pyproject.toml`, `README`, `environment.yml`. **Roll back:** this is the root — everything depends on it; don't revert.

### **3cd1f36** — `setuptools.build_meta` backend (2026-06-30)

**Effect:** Build backend switched for `setuptools≥61` compatibility. **Files:** `pyproject.toml`. **Roll back:** `git revert 3cd1f36` (only affects builds).

### **2e9f419** — add `launch.py` standalone launcher (2026-06-30)

**Effect:** `python /path/to/launch.py` runs the app from anywhere. **Files:** `launch.py`. **Roll back:** safe to revert; later hardened by `916ece2`.

### **fc754d0** — Linux style fix for invisible files/folders (2026-06-30)

**Effect:** Stylesheet fix so file/folder text is visible on Linux. **Files:** `midas_gui/style.py`. **Roll back:** `git revert fc754d0` (cosmetic).

### **acf0866** — Data Viewer: radial plot, ring picking, zoom persistence (2026-06-30)

**Effect:** Added the radial-integration plot, beam-centre ring picking, and zoom-persistence across a stack in the Data Viewer; widened left panels. **Files:** `tab_view.py` (+1-line width bumps in other tabs). **Roll back:** revert to remove the Data Viewer radial plot/ring picking.

### **d9e527e** — preserve zoom/pan across frames (2026-06-30)

**Effect:** Disabled `pyqtgraph autoRange` on redisplay so zoom/pan survive frame changes. **Files:** `widgets.py`. **Roll back:** revert if you *want* auto-refit per frame.

### **d50b588** — Data Viewer: intensity mask, calib loading, click-to-draw ring (2026-07-01)

**Effect:** 99.9-pct intensity-range mask default; load calibration (`json/poni/paramstest`); click-to-draw ring; test-data defaults point at local `test_data/`. **Files:** `tab_view.py`, `widgets.py`, `helpers.py`, `constants.py`, `tab_mask.py`. **Roll back:** revert to drop these Data Viewer features (note `helpers.py` geometry parsing added here is reused later — see `b381d8d`).

### **916ece2** — startup crash diagnostics + robust launcher (2026-07-01)

**Effect:** Logs uncaught exceptions/native faults to `~/midas_gui_error.log`, prevents PyQt slot-exception aborts, isolates failing tabs, hardens the launcher. **Files:** `app.py`, `launch.py`. **Roll back:** revert only if you want raw exceptions to propagate; not recommended.

### 7d010d7 — Abort buttons + dark/bright/background + auto-load (2026-07-01)

**Effect:** Abort on calibrate/batch; dark/bright/background field correction (file/folder/HDF5, index-range averaging, divide/subtract); browse auto-loads, "Load" buttons removed; compact FieldSelector. **Files:** `tab_batch/calibrate/corrections/mask/pdf/refine/texture/view.py`, `widgets.py`, `workers.py`, `helpers.py`. **Roll back:** wide-reaching; reverting removes field correction + auto-load across tabs. The Abort behaviour here was later re-worked by 399f3d7 (no `terminate()`).

### e862135 — MIDAS-style packaging (2026-07-01)

**Effect:** BSD-3 LICENSE, MIDAS-convention `pyproject.toml`, `release.sh` + `RELEASING.md`, headless tests/ smoke suite. **Files:** `LICENSE`, `README.md`, `RELEASING.md`, `pyproject.toml`, `release.sh`, `tests/`. **Roll back:** revert affects packaging/tests only.

### 41f28f5 — compact lattice, pixel-weighted azimuthal mean, readable menus (2026-07-06)

**Effect:** Data Viewer 3-per-row lattice + cubic quick-set; intensity-mask default max(99.99pct, 1e5); **pixel-weighted azimuthal mean** (selectable vs legacy  $\eta$ -bin mean) fixing off-detector-BC distortion; batch calibration accepts paramtest/poni/ json; readable right-click menus; no-scroll combos. **Files:** `helpers.py`, `workers.py`, `tab_view/batch/calibrate/pdf/refine/texture.py`, `widgets.py`, `style.py`. **Roll back:** to undo the azimuthal-averaging change specifically, prefer editing the `weighted` default rather than reverting the whole commit (it bundles UI changes).

### cd05ced — docs: not-on-PyPI install instructions (2026-07-06)

**Effect:** README clone+conda+run instructions; relative env self-install. **Files:** `README.md`, `environment.yml`. **Roll back:** docs only.

### 2e5f7c9 — docs: add gui\_documentation.md (2026-07-06)

**Effect:** First committed user guide. **Files:** `documentation/gui_documentation.md`. **Roll back:** docs only.

### 75c135c — docs: add gui\_documentation.pdf (2026-07-06)

**Effect:** PDF build of the guide. **Files:** `documentation/gui_documentation.pdf`. **Roll back:** docs only.

### aceb40f — docs: install via pip install midas\_suite (2026-07-06)

**Effect:** Corrected backend-install instructions. **Files:** `README.md`, `environment.yml`. **Roll back:** docs only.

### aa9395e — Unified Data-Loader panel + 3-panel splitter (tabs 0/2/3/4) (2026-07-07)

**Effect:** Shared `DataLoaderPanel` (Data/Dark/Bright/Background/Mask, per-mode frame controls) + `MaskSelector`; corrections applied on tabs 0/2/3/4; each of those tabs restructured into a draggable [Data Loader | Parameters | Display] splitter. **Files:** `widgets.py` (+442), `tab_view/calibrate/batch/refine.py` (large rewrites), `app.py`, `helpers.py`, `workers.py`, docs. **Roll back: high impact** — this is the base for the Batch/Calibrate/Refine and later Pump Probe layouts. Reverting breaks the 3-panel tabs and the Pump Probe tab's left panel. Avoid a blanket revert; fix forward instead.

## **b381d8d — Tab 6 polyatomic midas\_pdf pipeline + calibration loading (2026-07-08)**

**Effect:** PDF tab rewritten to the total-scattering pipeline; **vendored** `midas_pdf` under `midas_gui/_vendor/` + `pdf_backend.py`; helpers `geometry_fields_from_file` / `result_ns_from_geometry_file` (paramtest/json/poni) used by Calibrate and PDF "Load calibration file...". **Files:** `_vendor/midas_pdf/**` (new, large), `pdf_backend.py`, `tab_pdf.py`, `workers.py`, `helpers.py`, `tab_calibrate.py`, `constants.py`, `pyproject.toml`, docs. **Roll back:** reverting removes the PDF pipeline **and** the shared geometry-file loaders that later tabs (incl. Pump Probe via `spec_from_geometry_file`) rely on — revert with care.

## **1149afc — Batch live MONITOR + legend; PDF real-data default (2026-07-08)**

**Effect:** MONITOR button (stream mode) watches a folder and integrates new frames via `FolderMonitorWorker`, reusing a cached detector map (factored `build_integration_context()` + `write_profile()`, shared with `BatchWorker`); per-file legend; PDF tab defaults to real I(Q). **Files:** `tab_batch.py`, `workers.py`, `widgets.py`, `constants.py`, `tab_pdf.py`, docs. **Roll back:** to remove MONITOR only, revert this commit — but `build_integration_context()` introduced here is **reused by the Pump Probe worker** (590b410), so a full revert would break Pump Probe integration.

## **524f5f1 — Batch Clear results + inline labels; implementation-details doc (2026-07-09)**

**Effect:** "Clear results" button; inline per-file curve labels + line-width/font toolbar on stacked profiles; new `implementation_details.md/.pdf`. **Files:** `tab_batch.py`, `widgets.py`, `documentation/implementation_details.*`. **Roll back:** safe, localized to Batch + the new doc.

## **10df828 — Mask unbounded $\sigma$ + split auto-mask; Viewer↔Calibrate geometry (2026-07-09)**

**Effect:** Removed  $\sigma$  upper limits; split statistical auto-mask into independent Spatial-outlier / Temporal-constancy checkboxes; geometry hand-off `DataViewer.pushGeometry` / `Calibrate.pullGeometry` → `apply_geometry`. **Files:** `tab_mask.py`, `tab_view.py`, `tab_calibrate.py`, `app.py`, `workers.py`, docs. **Roll back:** revert to restore bounded  $\sigma$  / combined auto-mask / no geometry hand-off.

## **d135c80 — Batch stacked-profiles: themes, point+line, x-units, grid (2026-07-09)**

**Effect:** `StackedProfileViewer` publication/dark themes, point+line, symbol-size, R/2 $\theta$ /Q x-unit selector, grid toggle. (This viewer's styling is the template the Pump Probe plots later mirror.) **Files:** `widgets.py` (+228), `tab_batch.py`, docs. **Roll back:** localized to the Batch stacked-profile viewer.

## **2385f75 — Batch waterfall R/2 $\theta$ /Q axis; Mask 3-D HDF5 stack (2026-07-10)**

**Effect:** Waterfall x-unit selector via a converting `AxisItem` + `_convert_radial()`; Mask temporal-constancy reads a single 3-D (time,y,x) HDF5. **Files:** `widgets.py`, `tab_batch.py`, `tab_mask.py`, `workers.py`, docs. **Roll back:** `_convert_radial` / `_UnitAxis` introduced here are reused by the Pump Probe heatmap — don't fully revert if Pump Probe is present.

## **b628446 — clickable $\lambda$ label with K-edge foil menu (2026-07-10)**

**Effect:** Compact clickable " $\lambda$ " label → K-edge foils menu (sets  $\lambda$  from edge energy). `constants.K_EDGE_FOILS`, `HC_KEV_A`, `helpers.make_kedge_label`. **Files:** `constants.py`,

helpers.py, tab\_view/calibrate/pdf.py, docs. **Roll back:** localized; make\_pixel\_label (next) reuses the factored helper.

### 108ea7d — Data Viewer projection Skip-frames + compact fields (2026-07-10)

**Effect:** Projection "Skip frames" field; narrower Axis/Skip/pixel fields; Lsd thousands-separators. **Files:** tab\_view.py, docs. **Roll back:** localized.

### 3248638 — clickable px label with detector pixel-size menu (2026-07-10)

**Effect:** Clickable "px" label → GE/Varex/Pilatus/Eiger sizes; constants.PIXEL\_PRESETS; factored helpers.\_clickable\_menu\_label. **Files:** constants.py, helpers.py, tab\_view.py, tab\_calibrate.py, docs. **Roll back:** localized.

### 2b7a695 — Data Viewer intensity-statistics panel + histogram (2026-07-10)

**Effect:** IntensityStatsPanel (histogram + p70/p90/p99/p99.9/p99.99 with pixel counts) pinned to the loader bottom (stack mode); Current/All/projected scope. **Files:** widgets.py, tab\_view.py, docs. **Roll back:** localized.

### d30be2c — per-user configuration system + Preferences dialog (2026-07-12)

**Effect:** Per-user JSON config overlays shipped defaults at import; settings.py; prefs\_dialog.py (Geometry/Paths/Materials/Calibrants/Menus/Algorithms); Settings menu; DEFAULT\_KERNEL/PIPELINE/OUTPUT\_FORMAT/ERROR\_MODEL/COLORMAP; tab combos honor configured defaults. New config\_gui.md, config.example.json, tests/test\_config.py. **Files:** settings.py (new), prefs\_dialog.py (new), constants.py (+165), app.py, tab\_batch.py, tab\_calibrate.py, widgets.py, docs, tests. **Roll back: high impact** — the config overlay + DEFAULT\_\* and the Preferences dialog are extended by the modular-tabs commit ( 590b410 ). Reverting this would also break ui.visible\_tabs and the Tabs preferences section.

### 399f3d7 — Stability: clean worker shutdown, no terminate(), stdout guard (2026-07-12)

**Effect:** closeEvent stops all tab QThreads; removed every QThread.terminate() (cooperative interrupt + orphan instead); CalibrationWorker restores stdout only if still its own; Data Viewer projection moved to ProjectionWorker (off GUI thread). **Files:** app.py, tab\_batch.py, tab\_calibrate.py, tab\_view.py, workers.py. **Roll back:** reverting reintroduces the exit hard-crash risk and UI-freeze on projection — not recommended. Phase 1 of the 3-part review.

### 4462ee1 — Performance: waterfall O(N), debounce, caching (2026-07-12)

**Effect:** Waterfall pre-allocated buffer + 100 ms coalesced redraw (O(N) not O(N<sup>2</sup>)); 60 ms frame-slider debounce; cached radial bin-index grid; skip all-frame stats on frame-only changes. **Files:** tab\_view.py, widgets.py. **Roll back:** reverting slows large scans / scrubbing; behaviour otherwise same. Phase 2 of the review.

### 8846619 — Consistency: dedupe maps/writer, align lattices (2026-07-12)

**Effect:** helpers.\_PARAMSTEST\_DISTORTION derived from constants.\_V2\_T0\_V1 (removed duplicated \_V2V1 dicts); shared helpers.write\_standalone\_paramstest used by Calibrate + Export; LaB6/Si



`_LATT/_LC` aligned to `MATERIALS`; texture colormap honors `DEFAULT_COLORMAP`. **Files:** `constants.py`, `helpers.py`, `tab_calibrate.py`, `tab_export.py`, `tab_texture.py`. **Roll back:** reverting reintroduces duplicated code paths; low functional risk. Phase 3 of the review.

### 590b410 — Modular tabs + Pump Probe (TR-XRD) tab + plot quality (2026-07-12)

**Effect:** (1) **Modular tab visibility** — Data Viewer/Mask/Calibrate/Batch always shown, the rest toggled in Settings ▶ Preferences ▶ Tabs ( `ui.visible_tabs`, applied live); `app.apply_tab_visibility()`. (2) **New Pump Probe tab** ( `tab_pumpprobe.py` ) — TRR filename pooling, MIDAS-engine integration via new `PumpProbeWorker` (reuses `build_integration_context / integrate_frame`),  $\Delta I(q, \text{delay})$ , three-panel layout, four pyqtgraph views; ships a converted MIDAS `paramtest` as the default calibration. (3) **Plot quality** — white publication theme, black axes/labels/titles, readable legends,  $\Delta I$  colorbar, aligned reference panel, shared draw-mode + line/point/font-size toolbar, bounded pan/zoom. **Files:** `tab_pumpprobe.py` (new), `workers.py` (+ `PumpProbeWorker`), `app.py`, `constants.py`, `prefs_dialog.py`, `tests/test_smoke.py`, docs ( `gui_documentation`, `config_gui`, `config.example.json` ). **Roll back:** to remove **only Pump Probe**, delete `tab_pumpprobe.py` and its wiring in `app.py / workers.py`; to remove **only modular tabs**, revert the `app.apply_tab_visibility / prefs_dialog` Tabs section and `constants` tab lists. A full `git revert 590b410` removes both features together. Depends on `aa9395e` (`DataLoaderPanel`), `1149afc` (`build_integration_context`), `2385f75` (`_convert_radial / _UnitAxis`), `d30be2c` (`config/prefs`).

### 6d6f3fb — docs: add development\_history.md/.pdf (2026-07-13)

**Effect:** Added this per-commit change log + change→commit index + rollback guide. **Files:** `documentation/development_history.md`, `documentation/development_history.pdf`. **Roll back:** docs only.

### d5fafd8 — UI scaling: user-adjustable whole-interface zoom (HiDPI / 4K) (2026-07-13)

**Effect:** Whole-application scale (layout + fonts) for HiDPI / 4K monitors. `constants.DEFAULT_UI_SCALE` (config `ui.ui_scale`); `app.main()` applies it via `QT_SCALE_FACTOR` **before** the `QApplication` is created (so fixed widths, splitters, pyqtgraph and stylesheet px all scale uniformly) + `AA_UseHighDpiPixmaps`. Manual control in **Preferences ▶ Display** (spin + 100/125/150/200 % presets) and **Settings ▶ Interface scaling...**; both persist `ui.ui_scale` and offer an immediate self-restart (`MainWindow.restart_app` via `QProcess`) since the factor is read only at startup. **Files:** `constants.py`, `app.py`, `prefs_dialog.py`, docs ( `gui_documentation`, `config_gui`, `config.example.json` ). **Roll back:** `git revert d5fafd8`. Self-contained (depends only on the config system `d30be2c`); reverting removes the scale control and the app renders at 1.0x. To keep the feature but disable it, set `ui.ui_scale` to 1.0.

### c4e1c12 — Display Lsd in mm (calculations & files stay in $\mu\text{m}$ ) (2026-07-13)

**Effect:** The sample-to-detector distance is entered/shown in **mm** in the Data Viewer, the Calibrate seed, and Preferences ▶ Geometry; conversion to/from  $\mu\text{m}$  happens only at the display boundary (`DataViewerTab._lsd_um()`,  $\times 1000$  on read,  $\div 1000$  on set). `get_geometry / apply_geometry / manual_seed / prefs _assemble` still emit  $\mu\text{m}$ ; the config key stays `lsd_um` ( $\mu\text{m}$ ); calibration-file writers ( `paramtest.txt`, `calibration.json` ) are unchanged (always  $\mu\text{m}$ ). Batch drift-status label → mm. **Files:** `tab_view.py`, `tab_calibrate.py`, `prefs_dialog.py`,



`tab_batch.py`, docs. **Roll back:** `git revert c4e1c12` to return every Lsd field to a  $\mu\text{m}$  display. Purely a display-layer change — no stored/file/calculation values are affected either way.

---

## Rollback recipes (common intents)

---

- **Undo the Pump Probe tab only:** `git revert 590b410` removes Pump Probe *and* modular tabs. To drop Pump Probe alone, hand-remove `midas_gui/tab_pumpprobe.py`, its import/construction/wiring in `app.py`, `PumpProbeWorker` in `workers.py`, and its entry in `constants.OPTIONAL_TABS / DEFAULT_VISIBLE_TABS`.
  - **Undo modular tabs (show all 10 fixed again):** remove the `apply_tab_visibility` logic in `app.py` (add all tabs directly), the Tabs section in `prefs_dialog.py`, and the tab-list constants — see the `590b410` diff for the exact hunks.
  - **Undo the config system:** `git revert d30be2c` — but first revert `590b410`'s config touches, since it extends the same `ui` overlay and Preferences dialog.
  - **Revert the azimuthal-averaging change:** don't blanket-revert `41f28f5` (bundled with UI); instead set the `weighted` default back to the legacy  $\eta$ -bin mean in the batch/pump workers.
  - **Go back to the pre-3-panel single-column tabs:** `aa9395e` is the pivot; a revert is large and cascades to later tabs — prefer fixing forward.
- 

Generated from `git log`. To refresh after new commits: `git log --reverse --stat --date=short` and append entries above.